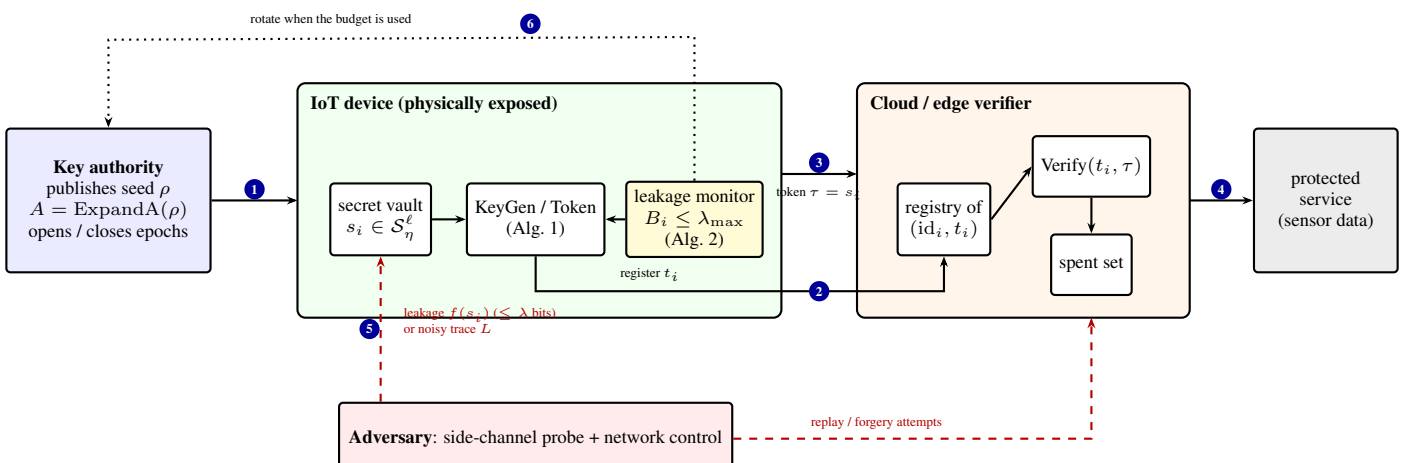

rotate when the budget is used
6
Key authority
publishes seed ρ
$A = \mathrm{ExpandA}(\rho)$
opens / closes epochs
1
IoT device (physically exposed)
secret vault
$s_i \in \mathcal{S}_\eta^\ell$
KeyGen / Token
(Alg. 1)
leakage monitor
$B_i \le \lambda_{\max}$
(Alg. 2)
register t_i
3
token $\tau = s_i$
Cloud / edge verifier
registry of
(id_i, t_i)
Verify(t_i, τ)
spent set
2
4
protected
service
(sensor data)
5
leakage $f(s_i)$ ($\le \lambda$ bits)
or noisy trace L
Adversary: side-channel probe + network control
replay / forgery attempts